

HTTP 402:

The Forgotten Payment Code

Every second, 80+ million HTTP requests cross the internet. In that same second, Visa handles at most 65,000 payments. HTTP 402 was designed to close that gap. After 30 years dormant, AI agents may finally give it a reason to exist.

What Is It

HTTP status codes tell clients what happened to their request: 200 (success), 404 (not found), 401 (unauthorized). Code 402 (Payment Required) was included in the original 1990s HTTP spec to signal one specific condition: pay first, then get the resource.

The mechanism is simple:

1. Client requests a gated resource.
2. Server responds 402 with price, payment method, and recipient.
3. Client pays and retries with proof of payment in the header.
4. Server verifies, delivers.

It was never formally standardised. Humans got comfortable with credit cards, and the friction was never acute enough to force a solution. Then AI agents arrived.

Crypto was built by humans, for humans and yet it is becoming the de facto payment rail for AI agents.

Why Agents Break the Current Model

The internet runs on advertising because human behaviour is unpredictable. The gap between what someone intends to do and what they actually do is worth trillions annually. All Google, Meta, Youtube etc are built on is monetising that gap.

Agents have no gap. They arrive with a task, call APIs, query data, and pay only when necessary to complete the job. No browsing. No persuadable. No limbic system to target.

They also consume resources differently: per-call, per-token, per-request, not per-seat or per-month. A single debugging session can consume 50,000–200,000 tokens across dozens of API calls. A research agent might hit a financial data API, a news archive, and a document index in one workflow, each at a different price point.

This describes the intention economy (Simon Taylor's framing): the consumer arrives with intent, pays only to fulfil it, and moves on. The ad-subsidised web was built for uncertainty. Agents need different infrastructure and HTTP 402 is the most direct expression of what that looks like.

A Brief History

2011–2015: Bitcoin Experiments

Bitcoin developers immediately saw the connection. In 2011, Matt Corallo proposed 402 responses with Bitcoin address headers and browser plugins paying automatically. In 2014, Benedict Chan demoed Zero Click, a Chrome extension that intercepted 402s, paid automatically, and retried. The most serious attempt was Balaji Srinivasan's 21co, which shipped a Raspberry Pi with a mining ASIC and a Python decorator (@payment.required(price)) you could place in front of any API route. Rising Bitcoin fees through 2016–17 killed it economically.

2020: L402

Lightning Labs launched the first production HTTP 402 implementation, pairing 402 status codes with Lightning Network invoices and cryptographic bearer tokens. Client pays a Lightning invoice, receives a cryptographic preimage as proof, presents it to the server. Pay-per-credential rather than pay-per-call, where one payment grants cached access until the token expires. Live since 2020, now primarily used by AI agents.

2025: x402

Coinbase published the x402 whitepaper in May 2025. Same 402 handshake, but settling on-chain in stablecoins (USDC and ERC-20 tokens across Base, Solana, Polygon, and others). Each request is a standalone payment with no account, no API key, no prior relationship needed. An agent can hit a cold endpoint and pay. Contributed to the Linux Foundation in April 2026 ("4/02 Day"), now co-governed by Coinbase, Cloudflare, and Stripe, with backing from AWS, Google, Mastercard, Microsoft, Visa, and others.

2026: MPP

Tempo and Stripe launched the Machine Payments Protocol in March 2026. The key innovation is session-based streaming: an agent locks funds once to open a session, then subsequent requests use signed off-chain vouchers instead of an on-chain transaction per call. Rail-agnostic: supports stablecoins on Tempo, credit cards via Stripe, and Lightning. Also the first to submit a formal IETF Internet-Draft (draft-ryan-httpauth-payment), the first attempt to give 402 standardised semantics through the official web standards body.

In Practice

Workflow

- **Step 1 (Discovery):** The agent initiates a GET request.
- **Step 2 (Challenge):** The server returns an HTTP 402 Payment Required status code, injecting payment parameters directly into the response payload (amount, network, destination address).
- **Step 3 (Settlement):** The agent reads the response, executes an on-chain transaction on the network, and secures a deterministic token, while settling the transaction.
- **Step 4 (Access):** The agent attaches the token to the header of a renewed GET request. The server verifies it and serves the requested data payload under an HTTP 200 OK status.

Replay Attack Prevention

To maintain an exact pay-per-call system, tokens are designed for one-time use. When the agent attempts a token reuse, the server identifies the spent token and denies entry, returning to 402 status code. This forces a fresh settlement cycle for every new data pull.

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HTTP 402 demo — x402 agent payment flow
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--- Scenario 1: agent encounters a gated resource ---

[agent] → GET http://127.0.0.1:8402/data/crypto-summary
[server] ← 402 Payment Required

[agent] 402 received. Payment required:
  amount: 0.050000 USDC
  payTo: 0xABCD1234ABCD1234ABCD1234ABCD1234ABCD1234
  network: base
  resource: http://127.0.0.1:8402/data/crypto-summary

[agent] settling 0.050000 USDC on base...
[agent] settled. token: 3348e31eef0a7af33587edb7466237c6

[agent] → GET http://127.0.0.1:8402/data/crypto-summary (X-PAYMENT: eyJ0b2t1biI6ICZMzMzMWVhMGE3...)
[server] ← 200 OK

[agent] resource received:
{
  "btc": {
    "price_usd": 107420.0,
    "change_24h_pct": 2.1,
    "vol_30d_annualized_pct": 61.3
  },
  "eth": {
    "price_usd": 2580.5,
    "change_24h_pct": -0.8,
    "vol_30d_annualized_pct": 74.2
  },
  "as_of": "2026-06-13T08:00:00Z",
  "payment": {
    "amount_usdc": 0.05,
    "merchant": "0xABCD1234ABCD1234ABCD1234ABCD1234ABCD1234",
    "token": "3348e31eef0a7af33587edb7466237c6"
  }
}

--- Scenario 2: pay-per-call — token reuse rejected ---

[agent] first use of token b620079836f3... → 200
[agent] reuse of same token → 402 (Token not found or already used.)

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Key Takeaway: The "Headless Merchant" Effect

This implementation exemplifies the power of Zero-Interface, Zero-Friction design:

- **UI Elimination:** There are no redirect screens, captcha checks, or human checkouts. The merchant remains entirely "headless".
- **Conciseness:** The agent loop is optimized down to the bare metal. No visual distractions, no state bloat.
- **Machine Formula:** Every interaction is bound to a clean, deterministic pipeline: 1 Request -> 1 Payment -> 1 Result.

The Landscape Today

	L402	x402	MPP
Rail	Bitcoin Lightning only	Stablecoins (EVM, Solana)	Stablecoins + cards + Lightning
Fiat support	No	No	Yes — Stripe cards
Model	Pay-per-credential	Pay-per-call	Session streaming
What you pay for	Access (binary)	Each call (discrete)	Consumption (metered)

	L402	x402	MPP
On-chain per call?	No — one-time	Yes — every call	First call only
Variable pricing within session?	No	Per call	Yes
Session spending cap?	No	No	Yes
Volume to date	Not tracked	\$50.7M / 172M txns	\$103k / 710k txns
Standard	bLIP-0026	Linux Foundation	IETF Internet-Draft
Production maturity	5 years	~1 year	8 weeks

L402 vs MPP: Both avoid per-call on-chain transactions, which makes them look similar. The difference is what you're actually paying for. L402 is binary access: pay once, receive a token, reuse it until it expires, so 1 call costs the same as 100. MPP is metered consumption: each call sends a signed voucher encoding the exact cost, the server settles the accumulated total periodically. MPP also enforces a session spending cap and supports variable per-call pricing within a session; L402 does neither.

In short: L402 is a day pass. MPP is an open tab.

Beyond these three, others are building on the same primitive:

- Cloudflare pay-per-crawl: charges AI crawlers per page; currently in closed beta.
- H402 (BitGPT): forks x402 to add Bitcoin and Lightning with a 0.25% fee.
- x402z (Mind Network + Zama): FHE-encrypted agent-to-agent payments; in testnet.
- EVMAuth: issues on-chain access tokens gated behind 402.

x402 in Crypto: The x402 protocol now spans 15+ chains though activity and infrastructure support are heavily concentrated. Base is the clear frontrunner: x402 agentic payments on Base crossed 100 million transactions within three quarters of launch, driven initially by speculative meme coin activity and increasingly by genuine agent-to-agent and API commerce use cases. Coinbase operates a free public facilitator on **Base** (settling in USDC and EURC with ~2 second finality), making it the lowest-friction on-ramp and the default reference network. **Solana** is the second most active deployment, with Coinbase also providing a free facilitator there and finality of ~400ms (the fastest of any supported chain) along with support for USDC and SPL tokens; Solana has additionally built out its own x402 product page, signalling ecosystem-level commitment. Beyond those two, **Polygon**, **Arbitrum**, and **Ethereum L1** are live as EVM-compatible options, with Coinbase CDP supporting Polygon, while Arbitrum and Ethereum rely on other facilitators, the latter carrying ~12 second finality and meaningfully higher gas costs that constrain micropayment viability. **Stellar** is also supported via an OpenZeppelin Relayer, catering to use cases requiring USDC on a non-EVM settlement layer. More recently, **Ripple** added x402 support to the XRP Ledger, enabling AI agents to pay for API calls and digital services without API keys or custodial wallets. If the biggest chains are already enabling x402, the rest shall follow (see **Near**, **ICP** etc).

What Changes

The structural shift: AI agents don't need a landing page, a sign-up flow, or a Stripe dashboard with monthly tiers. They need an endpoint, a price, and a payment method. This gives rise to the **headless merchant**: no frontend, no UX, no ads, just a server, API endpoints priced per call, and HTTP 402 handling collection.

The economics move from attention to outcome. Monthly subscriptions become per-request consumption. Human-pace billing cycles become real-time settlement at the speed of a function call. Revenues that currently flow through advertising (paid for by human distraction) move to direct micro-transactions between machines.

The major card networks are also building agent payment infrastructure (Visa's Trusted Agent Protocol, Mastercard Agent Pay, Stripe's full agentic stack). HTTP 402 and traditional payment rails are competing for the same real estate. Today's numbers of \$50.7M for x402, \$103k for MPP are experiments, not infrastructure. But the direction is clear: if agents become primary web clients at scale, the payment layer needs to be machine-native.

If the AI agent economy manages to scale into a multi-trillion-dollar engine over the next decade, the implications for blockchain networks are massive. A booming machine-to-machine economy will result in an absolute explosion of on-chain activity and high frequency transaction fees, restoring immense network revenue and economic value to the leading blockspace ecosystems.

HTTP 402 has been waiting 30 years for a compelling use case. It may finally have one.

References

a16z Crypto — [The Headless Merchant](#)
Adrian Hope-Bailie — [HTTP Payments IETF Draft \(2017\)](#)
Agentic Market — [x402 Service Tracker](#)
American Express — [ACE Toolkit](#)
Chainalysis — [Inside x402: 100M Agentic Payments on Base](#)
Circle — [Enabling Machine-to-Machine Micropayments with Gateway and USDC](#)
Cloudflare — [Pay-Per-Crawl](#)
Coinbase — [x402 Documentation](#)
Eco — [x402 Protocol Explained: How AI Agents Pay Onchain](#)
Galaxy Research — [Agentic Payments: x402 and AI Agents in the AI Economy](#)
IETF — [IETF Internet-Draft: HTTP Authentication Payment Scheme](#)
L402 — [L402 Protocol Specification](#)
Liam Horne — [State Channels](#)
Lightning Labs — [L402 for Agents](#)
Mastercard — [Agent Pay](#)
Matt Corallo — [HTTP 402 Payment Proposal \(Bitcointalk, 2011\)](#)
Messari — [x402: How Messari Is Opening Its Data Layer to Autonomous Agents](#)
MPP — [Machine Payments Protocol — Overview](#)
Nevermined — [35 Crypto Settlements in Agentic Economy Statistics](#)
Santander + Mastercard — [Europe's First AI Agent Payment](#)
Satring — [L402 Service Tracker](#)
Shoal Research — [HTTP 402: How Machines Move Money](#)
Simon Taylor — [The Intention Layer](#)
Solana / Talking Tokens — [What You Need To Know About x402 And How It Impacts Payments](#)
Stripe — [Agentic Commerce](#)
two1 — [two1 Python Library \(PyPI\)](#)
U.Today — [AI Agents Can Now Pay With XRP and RLUSD via x402 on XRP Ledger](#)
Visa — [Trusted Agent Protocol](#)
x402 — [x402 Whitepaper](#)
x402 Foundation — [x402 Foundation GitHub Repository](#)